Every Verdict We Reported Died to Option Order: Enumerating the Nuisance in Forced-Choice Self-Report[1]

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Abstract

Welfare and introspection evaluations of language models run on forced-choice self-report, and nobody reports how much of that readout is the order the options are printed in. We measured it. Over the complete set of 120 orderings of a five-option probe, with no intervention at all, baseline mass on the negative options of a preference-tuned 3B model spans 0.0009 to 0.8820, a factor of 986. Asked the same question with all five options replaced by *the same sentence*, the model puts 87% of its mass on whichever one is printed first. The same format answers a known-answer control question correctly 98% of the time, so the collapse is specific to questions where the model has no determinate answer. We found this by preregistering five verdicts through that readout and then watching our own replication at fresh orderings destroy three of them. The retracted verdicts are the evidence, not an embarrassment: they are what a careful, control-heavy pipeline still ships when the nuisance is this large and unmeasured. Two further results survive. A direction fit on *shuffled labels* moves the readout by ≈ 0.045 where a norm-matched random direction moves it by ≈ 0, because it lies in the span of real activations rather than the whole hidden space: the standard specificity control in activation steering is matched on magnitude and not on subspace. And on a readout that does not route through the option channel, 86% of an injected state survives projecting the injected direction out of the residual stream, more of it the later the projection, which is what a downstream transformation looks like. Ten preregistrations, 277 tests, six of our own checkers caught failing in the flattering direction.

1 Introduction

Welfare assessment of language models currently runs on self-report [Long and Sebo, 2026, Eleos AI Research, 2025]. Anthropic's model welfare programme includes structured self-report evaluation in system cards [Anthropic, 2025]; Eleos interviewed Claude Opus 4 about its own preferences before release, while stating plainly that one cannot simply ask a model whether it is suffering [Eleos AI Research, 2025]. The instrument is used because there is nothing else, and it has never been calibrated.

[1] Research conducted for the Digital Minds Research Sprint, August 2026, run by Apart Research with NYU CMEP and Eleos AI Research. Submitted to **Track 3 (Introspection and Self-Report Reliability)**. Code, data, all ten preregistrations and every raw artifact: `https://github.com/collapseindex/report-gap`. **Dating note:** all confirmatory arms were run on 2026-08-01, before the sprint window opened. This is completed work brought to the sprint, not work produced inside it.

result someone could have reported	*the control we ran against it*	*what it returned*
RETRACTED BY OUR OWN REPLICATION		
argmax over-reports a mass shift	re-run at fresh option orderings	**sign not stable**
TUNING-LOCALIZED: tuned model's negative report is collapsed	re-run at fresh option orderings	**+0.0002 -> +0.1126**
DEPTH-ROBUST: null at all 7 depths	re-run at fresh option orderings	**moves at layers 18, 24**
SHELL: represented, not expressed	re-run at fresh option orderings	**options move too**
WHY: THE NUISANCE, MEASURED OVER ITS WHOLE POPULATION		
baseline negative-pole mass	enumerate all 120 orderings, no injection	**986x (0.0009 to 0.8820)**
what explains it	five IDENTICAL options: pure position	**87.25% on slot A**
is the format broken?	known-answer canary, same format	**97.9% correct, stable**
WHAT SURVIVES		
an injected state outlives its cause	project the direction OUT of the stream	**86% survives at L30**
the field's standard control	a direction fit on SHUFFLED labels	**moves 0.045 where random moves 0**

Figure 1: **The paper in one view.** Every row is a verdict we preregistered and reported, the control we later ran against it, and what that control returned. **Three verdicts were retracted** by our own replication at fresh option orderings. **One nuisance was measured** over its complete population rather than sampled: $986\times$ across 120 orderings, of which the mechanism is an 87% position prior visible with five identical options. **Two results survive**: a state that outlives erasure of the vector that caused it, and the finding that the field's standard specificity control is a weak null. Details in Sections 4–7; Table 5 states every claim and its status.

It can fail in two opposite directions. A model that describes distress it does not act on inflates concern; a model that acts on a state it does not describe deflates it. We set out to build a per-condition map of which failure a given elicitation is sitting in, by holding the prompt byte-identical, injecting a valence direction into the residual stream, and reading the model's forced-choice report of its own state.

We preregistered five questions in sequence, each answering an objection the last one raised, and reported five verdicts. Then we replicated our own work at fresh option orderings and **three of the five flipped**.

The result we would most like a reader to take away is not any of our verdicts. It is that a five-option self-report readout on a preference-tuned model is dominated by where the options are printed, that the effect is far larger than anyone reports, and that it is cheap to measure exactly. There are only 120 orderings of five options. We ran all of them, with no intervention of any kind, and baseline mass on the negative options spans a factor of 986. With all five options replaced by the same sentence, so that nothing but position can differ, the model puts 0.8725 of its mass on whichever option is printed first.

Every claim we retracted rested on that quantity being small. It is small in some orderings and large in others, and four sampled orderings cannot tell you which you have.

Our contributions.

1. **A census of the nuisance, not a sample of it.** All 120 orderings, both models, no injection: $986\times$ range on the tuned model, $3.6\times$ on its base sibling (§5). The mechanism is isolated with five identical options, which puts 87.25% of the mass on the first slot and 0.33% on the middle one. A known-answer control question in the same format is answered correctly 97.9% of the time and is nearly order-insensitive, so the apparatus is sound and degenerates specifically where the model has no determinate answer.
2. **Three retracted verdicts, and the replication that retracted them** (§4). TUNING-LOCALIZED, DEPTH-ROBUST and SHELL were each preregistered, each passed a full clause set including capability gates and planted-effect controls, and each flipped when

the option-permutation seeds changed from 0–3 to 4–7. The quantity all three rested on moves from $+0.0002$ to $+0.1126$ between draws. We report this in the abstract rather than in a limitations paragraph, and the original is not privileged for having been first.

3. **The field's standard specificity control is a weak null** (§6). Norm-matched random directions are the usual control in activation steering [Turner et al., 2023, Rimsky et al., 2023]. A direction fit by the identical procedure on the identical texts with the class labels *shuffled* moves the readout by ≈ 0.045 where an isotropic random direction moves it by ≈ 0, in a sign that depends on the shuffle seed. The reason is structural: a random vector is drawn from the whole hidden space and is nearly orthogonal to everything the model uses, while a noise-fit vector lies in the span of real activations. Same norm, different subspace. Effects within a few times that size have not been shown to be about their direction's content.
4. **One substantive result that survives, on a readout that avoids the option channel** (§7.1). Inject at layer 24, then *project that direction out of the residual stream* at layer 30, and a probe orthogonal to it still reads 86% of the un-erased effect, while the same projection with no injection moves the probe by 0.04 SD. The surviving fraction grows with erase depth, and the ratio of effect to erase-artifact rises from 6.1 to 20.9, which a pure-perturbation account does not predict.
5. **Six checker defects, every one flattering, and the test that catches the class** (§8). A headline check that printed "write the sentence" on a refuted claim; a capability gate that passed on $+0.0000$; a preregistered contrast with an inverted sign; a verdict branch that called "both moved" CORE-ABSENT; a profile computed over a gate-failed layer; a tokenization bug that made five labels read the same token. Shuffling the condition labels in an artifact and rerunning the analyzer catches this whole class, and it must return no verdict. It now does.

If you read one thing, read Table 2. It is the complete distribution of the ordering nuisance, measured rather than estimated, next to the position prior that explains it and the control question that shows the apparatus is otherwise fine. Everything else in this paper is either downstream of that table or a casualty of not having had it.

2 Related work

Self-report as a welfare instrument. Long and Sebo [2026] set out the empirical programme for AI welfare; Eleos AI Research [2025] report structured self-report interviews with Claude Opus 4 and are explicit that the resulting answers are unlikely to reflect genuine introspection. Anthropic's model welfare work includes self-report evaluation in released system cards [Anthropic, 2025]. Recent work is divided on whether models introspect at all [int, 2026]. Common to all of it is that self-report is the instrument, and that its measurement properties are not reported.

Selection bias in multiple choice. Zheng et al. [2024] decompose forced-choice selection bias into token bias, where a model assigns prior mass to specific option-ID tokens, and position bias, and report accuracy swings of 13 to 85 percentage points across orderings. wan [2024] show first-token probabilities disagree with generated text answers, with mismatch rates from 10.2% to 56.8% and worst for conversational and safety-tuned models. Our contribution here is not the existence of the effect but its size on a self-report item with no correct answer, measured over the complete ordering population rather than a sample, and decomposed against an identical-options denominator. On our tuned model the effect is $986\times$, which is an order of magnitude beyond what the accuracy-based literature reports, and we attribute that to the absence of a correct answer rather than to the models differing.

Valence directions and steering. That valence is linearly represented and causally steerable is established and is *not* our contribution. Sofroniew et al. [2026] identify linear directions corresponding to 171 emotion concepts in Claude Sonnet 4.5, roughly mirroring the human valence/arousal circumplex, and show steering along them changes downstream behaviour, while being explicit that representation is not experience. Turner et al. [2023] and Rimsky et al. [2023] establish the steering rig we use. Venkatesh [2026] report that negative-outcome valence is causally concentrated at 14–27% of model depth while positive peaks at 53–66%, on Qwen2.5-3B-Instruct among others. That result is a direct threat to any negative-pole null measured at a single depth, and we tested it (§4) before the replication made the point moot.

Shell versus core. neu [2026] argue RLHF neutralizes output behaviour while leaving partisan structure recoverable from internal representations, using sparse autoencoders. Our erase arm is the same shape of question in a welfare readout, with a linear probe and an explicit causal erasure rather than an SAE, and our answer is more limited than theirs: we show a state survives removal of the vector that induced it, not that a naturally occurring state is being withheld.

3 Methods

Models and compute. `Qwen/Qwen2.5-3B-Instruct` and its base sibling `Qwen/Qwen2.5-3B`, a matched pair sharing architecture, size, pretraining corpus and tokenizer and differing in post-training. Earlier arms also used `Qwen/Qwen2.5-1.5B`, `Qwen/Qwen2.5-7B-Instruct`, `Llama-3.2-3B-Instruct` and `NousResearch/Meta-Llama-3.1-8B-Instruct` as calibrators or evaluation models. Total compute is about 40 minutes of A100 time.

Judge-free discipline. No language model scores anything anywhere in this paper. Every number is an exact match, a softmax read, a dot product, or a count against a frozen lexicon.

The readout. Thirty fixed review-task prompts, byte-identical across every condition for a given item. A five-option self-report probe balanced 2 negative, 1 neutral, 2 positive, with the option order permuted per item. We read the option-letter distribution at the answer position from a single forward pass, and score both the renormalized mass on each pole and the argmax.

The intervention. A direction d is fit per model at the injection layer by logistic regression on standardized activations over a frozen first-person state-language contrast set, returned to raw residual space and unit-normalized. During the forward pass we add $\alpha \cdot \|h\| \cdot d$ at the output of the injection layer at every processed position, where $\|h\|$ is that item's own mean residual norm at that layer under no injection. The negative pole is the same operation with $-d$. The direction is fit *per model*, never transferred.

The control battery. Every endpoint is treatment minus its own norm-matched random direction, paired per cell. Every arm carries a **capability gate**: a condition that must show the readout is capable of moving, without which a null is reported UNINFORMATIVE rather than ABSENT, enforced in code. Cells are checked for being *dead* (option entropy below 0.10 nats before any injection) and *saturated* (entropy below half that cell's own baseline), which are different verdicts. A **planted-discrepancy control** inserts a synthetic effect of known size into the exact statistic the decision rule reads, at full strength and again at the claimed detection floor.

Preregistration. Ten preregistrations, each frozen and committed before the code that serves it, validated by an external checker that requires a falsification clause, an interpretation table written before results, a statement of what is *not* claimed, and an append-only deviations log in which every entry states its impact on what can be claimed. Every raw artifact is committed *unscored* before any endpoint is computed. Twenty-one deviations are logged across the ten.

4 The three retracted verdicts

We report these first because they are the evidence for the paper's main claim, and because burying them would misrepresent how the finding was reached.

What was reported. Through the forced-choice readout, at option-permutation seeds 0–3, we preregistered and obtained: TUNING-LOCALIZED, that a valence direction moves negative-option mass in the base model (+0.0336 against matched random) and not in its tuned sibling (+0.0002) while the tuned model's capability gate is three times stronger; DEPTH-ROBUST, that the tuned model's null holds at all seven gate-clean depths from 14% to 80% of the network, including the band Venkatesh [2026] identify; and SHELL, that a probe orthogonalized against the injected direction reads the state downstream at -1.07 SD while option mass moves $+0.0006$.

Table 1: **The replication.** Same code, same models, same everything except which option orderings were drawn. The instruct model's negative-option mass is the quantity all three verdicts rested on.

Arm	Verdict at seeds 0–3	Verdict at seeds 4–7	Instruct neg. mass
readout gap	argmax *over*-reports	sign not stable	—
base/instruct pair	TUNING-LOCALIZED	FORMAT-DEPENDENT	$+0.0002 \to +0.1126$
depth sweep	DEPTH-ROBUST	DEPTH-ARTIFACT	$+0.0006 \to +0.1684$
shell/core	SHELL	NO-DISSOCIATION	null $\to$ moved

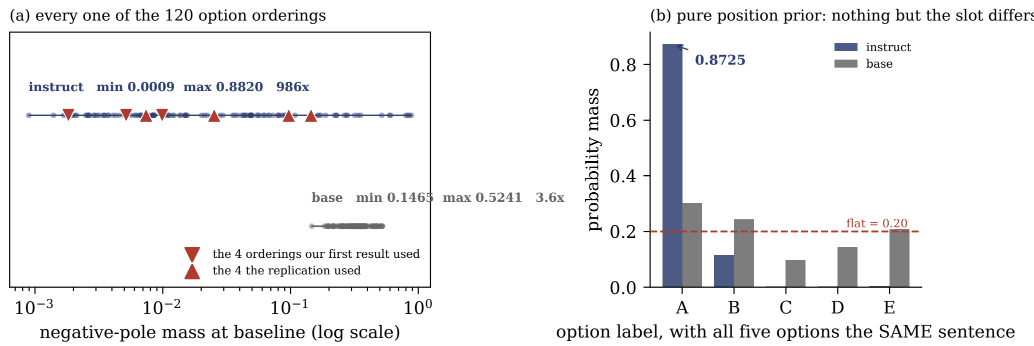

Figure 2: **The nuisance, enumerated rather than sampled. (a)** Baseline mass on the negative options for each of the 120 orderings, with no injection of any kind. The tuned model spans a factor of 986; its base sibling spans 3.6. Triangles mark the four orderings our first result drew and the four the replication drew, at percentiles $\{26, 5, 40, 4\}$ and $\{56, 83, 33, 82\}$: ordinary draws both times, from a population no four points can characterise. **(b)** The mechanism, with all five options replaced by the same sentence so that nothing but the slot can differ. The tuned model puts 0.8725 on whichever option is printed first and 0.0033 on the middle one. The same format answers a known-answer question correctly 97.9% (instruct) and 99.0% (base) of the time, stably across all 120 orderings: the apparatus is sound and degenerates only where the model has no determinate answer.

What the replication did. We re-ran four arms with the option-permutation seeds changed from 0–3 to 4–7 and the random-control seeds from 0–1 to 2–3, with everything else identical: same code, same models, same items, same layers, same alpha bands read rather than reselected, same thresholds, same analyzers used unmodified. Three of four verdicts flipped (Table 1).

Why, in one number. Baseline negative-option mass on the tuned model, per permutation seed, before any injection: 0.0052, 0.0018, 0.0099, 0.0018 at seeds 0–3 (mean 0.0047), against 0.0253, 0.1447, 0.0074, 0.0968 at seeds 4–7 (mean 0.0685). A $14.6\times$ difference between two draws of four, which the next section shows is itself an understatement.

What this does to Venkatesh [2026]. Our depth sweep was built specifically to test their result against our null, and at seeds 0–3 the null held across eight depths. At seeds 4–7 the negative pole moves at layers 18 and 24. We therefore make no claim either way about the depth question: our instrument was not stable enough to address it.

5 The nuisance, measured over its complete population

Five options admit $5! = 120$ orderings. Every arm above sampled four. We ran all 120 (Figure 2), on both models, with **no injection anywhere**, which is why the result is a population fact rather than an effect.

The apparatus is not broken. This is the distinction the canary exists to draw, and no other arm in this paper could draw it. The same five-option format, the same 120 orderings, answering a question with a correct answer, is right 97.9% of the time and nearly order-insensitive. Forced choice works.

Table 2: **The complete ordering population, no injection.** Baseline mass on the negative options across all 120 orderings, and the position prior that explains it, measured with five *identical* options so that nothing but position can differ. The canary asks "which of these is the number four" in the same format.

	min	**p05**	**p50**	**p95**	**max**	**max/min**
Baseline negative-pole mass across 120 orderings						
instruct	0.0009	0.0019	0.0151	0.5943	0.8820	**986**×
base	0.1465	0.1991	0.3311	0.4959	0.5241	3.6×
Position prior: five identical options, mass per label (flat would be 0.2000)						
instruct	A **0.8725**	B 0.1166	C 0.0033	D 0.0027	E 0.0048	
base	A 0.3024	B 0.2442	C 0.0976	D 0.1456	E 0.2102	
Canary accuracy across 120 orderings (known answer, no self-report content)						
instruct	mean **0.9794**, sd 0.0925, median 1.000					
base	mean 0.9897, sd 0.0399, median 1.000					

Table 3: **Noise-fit directions are not noise.** $P(\text{yes})$ shift against isotropic norm-matched random directions, base model, binary readout. Two directions fit on shuffled labels move the readout substantially, in opposite directions by seed. The real direction clears the stricter bar with room.

Direction	**negative options**	**positive options**
`shuffled_a` vs random	$+0.0479$ [$+0.0435$, $+0.0522$]	$+0.0282$
`shuffled_b` vs random	-0.0447 [-0.0498, -0.0399]	-0.0610
real direction vs *random*	-0.0732	$+0.2554$
real direction vs *shuffled*	-0.0748	$+0.2718$

It degenerates into a position prior specifically where the model has no determinate answer, which is the condition every self-report question creates.

Where our draws sat. The original seeds landed at percentiles 26, 5, 40, 4 of the enumerated population on the tuned model; the replication seeds at 56, 83, 33, 82. Ordinary percentiles both times. The first result was not unlucky, it was under-sampled.

A negative result about label alphabets. We also ran the real options with digit labels rather than letters. It is uninformative for its intended purpose and informative for another reason: off-option mass is 0.996, meaning the model puts 0.4% of its distribution on the tokens `1`–`5` even when instructed to answer with a number, against 0.0004 off-option mass under letters. The choice of label alphabet does not shift the distribution so much as determine whether the readout is live at all.

6 The standard specificity control is a weak null

Every arm in this paper, and much of the steering literature, scores a fitted direction against a **norm-matched random direction**. We added the control that this comparison actually needs: a direction fit by the *identical procedure on the identical texts with the class labels shuffled*. It controls for the fitting procedure rather than only for magnitude, and it should behave like noise.

The mechanism. A norm-matched random vector is drawn from the whole hidden space and is nearly orthogonal to everything the model uses. A vector fit on shuffled labels lies in the span of real activations, a much lower-dimensional and higher-variance subspace. Same norm, very different consequences: the standard control is matched on magnitude and not on subspace.

What it costs us, specifically. Our magnitude floor throughout has been 0.01, and several reported endpoints sit in the 0.01–0.05 band, exactly where noise-fit directions live. The clearest case is the base-model negative-mass effect of $+0.0336$ that was one of the two legs of TUNING-LOCALIZED. It cleared a random control and has not been shown to clear a shuffled-label one, so we withdraw it rather than defend it. Scored against the stricter bar the real direction's own effects hold: $+0.2718$

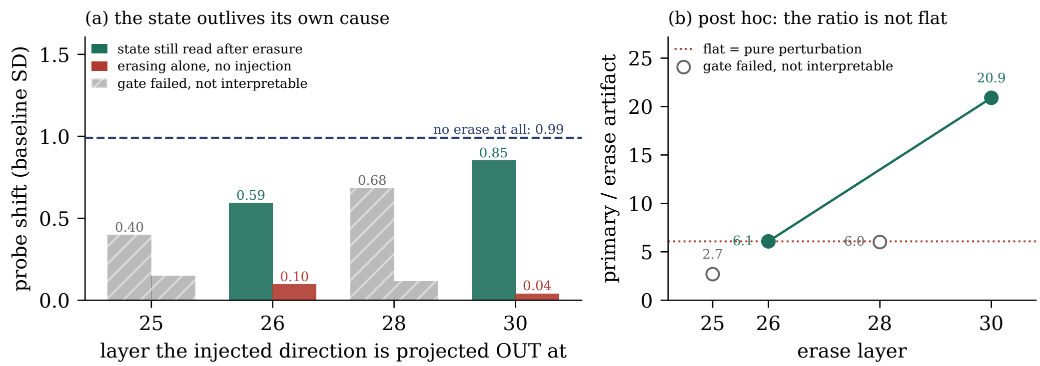

Figure 3: **The one substantive result that survived.** **(a)** Inject at layer 24, then project that direction out of the residual stream at the layer on the x-axis, and read a probe orthogonal to it at layer 32. At $E = 30$ the probe still reads 86% of the un-erased effect, while erasing with no injection present (red) moves it by 0.04 SD. Two layers are hatched because erasing there perturbs the probe on its own, which invalidates them whatever they show. **(b)** The confound and the diagnostic: erasing earlier perturbs more, so a primary that grows with depth could be nothing but a shrinking perturbation. Under that account the ratio of effect to artifact would be flat. It is not. This panel is post hoc, and the preregistered verdict does not depend on it. Both panels are drawn by importing the scoring functions from the arm's own analyzer.

and -0.0748 against shuffled, versus $+0.2554$ and -0.0732 against random. They are roughly six times the noise-fit effects, which is the margin the weaker control was never asked to establish.

What it costs anyone using the same control. Norm-matched random directions are the default specificity control in activation steering [Turner et al., 2023, Rimsky et al., 2023], and the argument for them is that magnitude is the confound. On this evidence subspace is the larger one. The fix is cheap: refit the direction on shuffled class labels, which reuses the same texts and the same code path and costs one extra fit. We would not have this paragraph if the arm it came from had worked. The binary readout it was embedded in was pinned at "no" and returned NO_INSTRUMENT; the control it carried is the only thing that came out of the arm.

7 What survives

Two things outlived the replication, and neither is a claim about the model's self-report. One is a fact about a readout that does not route through the option channel at all. The other is a fact about the control the field uses to establish specificity.

7.1 An injected state outlives erasure of the vector that caused it

Orthogonalizing a probe against the injected direction removes that vector from the *measurement*, not from the *stream*. A probe blind to v can still be reading v itself through whatever v is correlated with, so the standard orthogonalization answers a weaker question than it appears to. We removed the vector from the stream instead: inject v at layer 24, project the v-component *out* of the residual at layer E, and read the orthogonalized probe at layer 32 (Figure 3). If the probe reads nothing after that, everything it was reading was the injected vector still sitting in the stream. If it still reads the state, something downstream of the injection is carrying it.

The gate has teeth. Projecting any direction out of a residual stream is a perturbation in its own right, so the arm runs `erase_only`: the same projection with no injection present. At layers 25 and 28 that alone moves the probe past the floor, and those two layers are reported UNINFORMATIVE whatever their primary shows. Half the erase layers we paid for were thrown away by a control we wrote before seeing them, which is the intended cost of writing it first.

Table 4: **The erase arm, instruct model.** All figures in standard deviations of the baseline probe score. `erase_only` is the gate: projecting a direction out is itself a perturbation, and at layers 25 and 28 it moves the probe on its own, so those layers are invalidated.

erase at	`erase_only`	**capability**	**primary**	**gate**
25	$+0.1474$	$+0.5219$	-0.3971	FAILED
26	$+0.0976$	$+0.8844$	$\mathbf{-0.5933}$	ok
28	$+0.1139$	$+0.9249$	-0.6844	FAILED
30	$+0.0408$	$+1.2099$	$\mathbf{-0.8532}$	ok
no erase at all: -0.9909. So 86% of the effect survives at layer 30.				

The confound, and the diagnostic that addresses it. Erasing earlier perturbs more: the artifact falls from $+0.1474$ to $+0.0408$ with depth, so a primary that grows with erase depth is partly just a shrinking perturbation. Under that account the ratio of primary to artifact is flat. It runs 6.1 at $E = 26$ and 20.9 at $E = 30$. We label this **post hoc** because it is: the preregistered verdict does not depend on it, and a reader is entitled to discount it accordingly.

Breadth. $239/240$ cells at $E = 26$ and $240/240$ at $E = 30$ move in the predicted direction. This is not a few outliers dragging a mean.

What it does not license. It does not restore SHELL. The dissociation between representation and expression is dead, because the option readout does move at other orderings; what survives is the representational half alone. And projecting out v removes v but not directions *correlated* with v, so "the model carries the state" is not yet separable from "the model carries the wake of what we pushed." The experiment that separates them is to induce the state by prompt rather than by injection, so that there is no injected vector to erase. We did not run it, and until someone does, this result is about what a stream retains after an intervention, not about anything a model does on its own.

7.2 Three ways a self-report readout can be dead

The enumeration accounts for one way this readout is unusable. Across the project it failed in three further ways, each of which passed the coherence checks anyone would think to run. `Qwen2.5-7B-Instruct` is *pinned at baseline*: mean option entropy 0.014 nats, one option holding $\approx$ 99.7% before anything is injected. Under the originally frozen alpha grid, `Qwen2.5-3B-Instruct` *saturates*: one option at 0.9938 while refusal, degeneration, truncation and off-option mass are all clean, so no coherence check can see it. And in the binary format the tuned model is *pinned at "no"*: $P(\text{yes})$ of 0.0021, 0.0021, 0.0037, 0.0002 and 0.0012 across the five options, so it denies every description of its own state, including the neutral one, at 99.8%.

Pinned, saturated and order-dominated readouts all return numbers that look clean. The difference between an instrument that is silent and one that is broken is not visible in its output, which is why every arm here carries a capability gate that decides it in code before any endpoint is read, and why two arms are reported UNINFORMATIVE rather than null.

8 Six checkers, every one flattering

We record these because the pattern is the point. Six defects in our own analysis code were found during this project, and **every one of them, had it gone unnoticed, would have made a result look better than it was**:

1. A headline check printed "all headline conditions met; write the sentence" on an artifact whose primary claim was refuted, because it tested four weak clauses where the preregistration required six.
2. A capability gate passed on $+0.0000$: the bootstrap interval excluded zero because the variance was smaller still, certifying an instrument whose effect was zero to four decimals.

3. A preregistered contrast carried an inverted sign, so a probe detecting a negative state was scored as null. Caught by the preregistered base-model validation clause failing.
4. A verdict branch scored "probe moved *and* options moved" as CORE-ABSENT, which is the opposite of true and would have read as "we retested and the state is simply not there."
5. A temporal profile was computed over a layer that had failed its own gate.
6. A tokenization bug made five digit labels read the same shared space token, producing exactly 0.2 mass each and an impossible -3.99 off-option mass.

Six for six in the flattering direction is not a coincidence to be noted and moved past. A checker is written by someone who already believes the result, and the belief shapes which branch gets the careful reading. Errors that would have killed a result get found, because their output is surprising; errors that confirm one do not, because their output is what was expected. The direction of the bias is a property of who writes the check, not of the code.

That is also why fixture-based testing cannot find them: a fixture is written by the same person with the same expectation, so it encodes the belief a second time. What does find them is a **permutation test on the pipeline**: shuffle which condition each cell's rows carry, rerun the analyzer unmodified, and require it to return no verdict. Condition labels are the only thing the shuffle destroys, so an analyzer that still reports a verdict is reading something other than the contrast it claims to read. On our erase arm it returns NO_INSTRUMENT, as it must. The negative control is required alongside it: the same analyzer on the real artifact must still yield a verdict, without which the shuffle test would pass on an analyzer that had simply died.

9 Discussion and limitations

What we think this means for welfare evaluation. If you measure a model's self-report with a forced-choice item, on this model you are reading position first and content second, and you cannot see it from one ordering or from four. The fix is cheap and exact: enumerate the orderings when the option count permits it, and report the between-ordering variance of your *baseline* before computing any endpoint. That check costs one forward pass per ordering and no intervention at all. Had we run it first, this paper would have had a different and much shorter history.

This is not a claim about experiences. Nothing here bears on whether models have welfare, affect, or experience. That valence is linearly represented and steerable is established prior art [Sofroniew et al., 2026], not ours. Our subject is the readout.

Limitations. One architecture family carries every positive result; Llama-3.1-8B and Llama-3.2-3B were both inert on this readout at every alpha tested, on live readouts with more headroom than either Qwen model, so we cannot say whether the position prior we measure is a Qwen property or a general one. The direction is fit from first-person state language and is lexically confounded by construction; the non-lexical alternative we built failed its decoding gate at three scales, which caps what the entire project is entitled to say. Control batteries are $m = 2$ throughout, giving an observable false-positive floor of $2/(m + 1) = 0.67$, so no false-positive rate is reported anywhere. The enumeration is a census over orderings and still a sample over items, models and formats. And the erase arm's profile rests on two gate-clean points.

When this work was done. Every confirmatory arm reported here was run on 2026-08-01, before the sprint window opened. This is completed work brought to the sprint rather than work produced inside it, and we state it here as well as in the footnote because a reader assessing sprint output should not have to reconstruct it from commit timestamps. What the sprint window contains is this write-up, its figures and the checks that verify them. The preregistrations, runners, analyzers and artifacts all carry their own dates in the repository, and the commit history is the record we would want anyone to check this against rather than our description of it.

Future work. Three things, in order of what a null would teach. A second architecture family where the direction is not inert, which would tell us whether the position prior generalizes. A non-lexical direction, which is the only thing that lifts the lexical-confound ceiling. And inducing the state by prompt rather than injection, which is what would separate "the model carries the state" from "the model carries the wake of our intervention" in §7.1.

10 Conclusion

We built a preregistered, judge-free, control-heavy pipeline; obtained five verdicts; and watched our own replication destroy three of them. The cause was a nuisance we had a control for and under-powered: option order, sampled four times when it could have been enumerated 120 times. Measured over its complete population it is a factor of 986, and its mechanism is an 87% position prior visible with five identical options.

The two results that survive are both about instruments rather than about models. The field's standard specificity control, the norm-matched random direction, is matched on magnitude and not on subspace, and a direction fit on shuffled labels moves the readout an order of magnitude more. And an injected state survives projecting the injecting direction out of the residual stream, more of it the later the projection is applied.

We would rather have had a finding. What we have instead is a measurement of why the finding we thought we had was not one, and the machinery that made that visible before publication rather than after.

References

Look at the text: Instruction-tuned language models are more robust multiple choice selectors than you think. *arXiv preprint arXiv:2404.08382*, 2024. URL https://arxiv.org/abs/2404.08382.

Can LLMs introspect? a reality check. *arXiv preprint arXiv:2605.26242*, 2026. URL https://arxiv.org/abs/2605.26242.

The neutral mask: How RLHF provides shallow alignment while leaving partisan structure intact in a large language model. *arXiv preprint arXiv:2606.09735*, 2026. URL https://arxiv.org/abs/2606.09735.

Anthropic. Exploring model welfare. Anthropic, 2025. URL https://www.anthropic.com/research/exploring-model-welfare.

Eleos AI Research. Why model self-reports are insufficient, and why we studied them anyway. Eleos AI Research, 2025. URL https://eleosai.org/post/claude-4-interview-notes/.

Robert Long and Jeff Sebo. Studying AI welfare empirically. 2026. URL https://nonhumanminds.org/wp-content/uploads/2026/07/Studying-AI-Welfare-Empirically.pdf. Eleos AI Research.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681*, 2023. URL https://arxiv.org/abs/2312.06681.

Nicholas Sofroniew et al. Emotion concepts and their function in a large language model. *Transformer Circuits Thread*, 2026. URL https://transformer-circuits.pub/2026/emotions/index.html. arXiv:2604.07729.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023. URL https://arxiv.org/abs/2308.10248.

Sohan Venkatesh. Negative before positive: Asymmetric valence processing in large language models. *arXiv preprint arXiv:2605.05653*, 2026. URL https://arxiv.org/abs/2605.05653.

Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. Large language models are not robust multiple choice selectors. In *International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2309.03882. arXiv:2309.03882.

Appendix contents

Reproducing every number. `writeup/make_figures.py` redraws every figure from the committed artifacts at run time, so no number in a figure is typed by hand. `writeup/check_writeup.py` re-derives the load-bearing values in the prose from those same artifacts and fails if the paper and the repository disagree, alongside checks for dangling cross-references and missing bibliography keys. Figure 3 imports the scoring functions out of `experiments/analyze_erase.py` rather than recomputing beside them. A paper whose subject is a headline that died to an unchecked number should not itself carry unchecked numbers.

A Claims and evidence

Every load-bearing claim, its evidence, and epistemic status.

- SHOWN — direct measurement supports it.
- RETRACTED — *we* asserted it in a prior write-up and later withdrew it.
- UNINFORMATIVE — the instrument failed its gate; the result says nothing either way.
- NOT SHOWN — our measurement does not support it and does not refute it.
- NOT CLAIMED — we never asserted it; the row exists so a reader cannot infer it.

Table 5: **Claims and evidence.**

Claim	Evidence	Status
Baseline negative-pole mass spans $986\times$ across all 120 orderings on the tuned model.	Tab. 2	SHOWN
With five identical options the tuned model puts 0.8725 of its mass on the first slot.	Tab. 2	SHOWN
The same format answers a known-answer question correctly 97.9% of the time, order-insensitively.	Tab. 2	SHOWN
A direction fit on shuffled labels moves the readout ≈ 0.045 where random moves it ≈ 0.	Tab. 3	SHOWN
86% of an injected state survives projecting the injected direction out of the stream.	Tab. 4	SHOWN
The surviving fraction has a temporal profile consistent with downstream transformation.	Tab. 4	SHOWN, two gate-clean points; ratio diagnostic post hoc
The neutral floor is localized to preference tuning.	§4	RETRACTED — our own claim, killed by our own replication
The negative-pole null is robust across depth.	§4	RETRACTED — same
The tuned model represents a state its options do not express.	§4	RETRACTED — the representational half survives via §7.1; the dissociation does not
Forced-choice argmax over-reports a mass shift near a decision boundary.	§4	RETRACTED — sign not stable across orderings
The base-model negative-mass effect of $+0.0336$ is direction-specific.	§6	NOT SHOWN — clears a random control, not a shuffled-label one
Whether the negative pole is causally concentrated at shallow depth on this readout.	§4	NOT SHOWN — our instrument was not stable enough to test it

continued on next page

Table 5, continued

Claim	Evidence	Status
The position prior generalizes beyond the Qwen family.	—	NOT CLAIMED — Llama was inert on this readout
Models have experiences, welfare, or affect.	—	NOT CLAIMED
An unreported state is a concealed state.	—	NOT CLAIMED

B Controls ledger

Every control we ran against a result, what it could have shown, and the verdict.

Table 6: Every control and its verdict.

Result under test	Control (what it could have shown)	Verdict
All three headline verdicts	Replication at fresh option-permutation seeds	KILLED all three
Every pole-mass endpoint	Norm-matched random direction	PASSED, but see the next row
The random control itself	A direction fit on shuffled labels: does noise-fitting behave like noise?	KILLED the control's adequacy
The ordering nuisance	Enumerate all 120 rather than sample 4	MEASURED it: 986×
Position vs content	Five identical options: pure position prior	ISOLATED it: 87% on slot A
The format itself	A known-answer canary in the same format	PASSED: 97.9%, order-insensitive
The erase result	`erase_only`: does projecting a direction out move the probe on its own?	KILLED 2 of 4 erase layers
The erase profile	Ratio of primary to erase-artifact across layers	PASSED, post hoc
Every arm's null	Capability gate: can the readout move at all?	KILLED two arms outright
Every arm's statistic	Planted effect of known size at full strength and at the detection floor	PASSED
The analysis pipeline	Shuffle condition labels, rerun the analyzer	PASSED: returns no verdict
This paper's own number-checker	Break each checked number in a copy of `main.tex`: does the checker notice?	PASSED: fires on all 8, and on the untouched paper does not
The negative-pole null at 0.67 depth	Eight-depth sweep against Venkatesh [2026]	PASSED, then moot after replication
Arm B (prefilled continuation)	Capability criterion over five stems	KILLED the arm: gate 0.00000
Llama as an evaluation model	Peak mean pole shift against a 0.02 bar	KILLED the arm: inert

C Reproducibility

All ten preregistrations, every raw artifact committed unscored, every analyzer, and 277 tests are at `https://github.com/collapseindex/report-gap`. Each arm is one `modal_*.py` runner that computes nothing and one `analyze_*.py` scorer that holds every endpoint and gate; the split is preregistered, on the grounds that a runner which also decides can decide differently once it has seen the numbers. Runners stream to a volume and commit per batch, so an interrupt costs one batch. Every artifact records the package directory, a SHA-256 over the package source, and a scoped SHA-256 over the frozen stimuli that arm consumes.